



The University of Texas at Austin
McCombs School of Business

Airbnb Pricing Improvement

Predicting listing price with linear and polynomial regression

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Machine Learning · March 2024

Post Graduate Program in AI & ML: Business Applications

McCombs School of Business, The University of Texas at Austin

PROJECT

02

Machine Learning

Regression models predicting listing price from 74,111 Airbnb records; pricing strategy derived from the coefficients.

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Executive summary — recommendations

01

Collect the missing drivers

Location, check-out policy, amenities, seasonal or event pricing and occupancy are absent from the data. Adding them is the fastest route to a better model.

02

Lean on entire-home listings

Entire homes and apartments command the highest prices. Converting or upgrading space to whole-property listings raises revenue ceiling.

03

Invest in bathrooms

Bathroom count carries the largest positive coefficient among numeric features. Assess renovation and addition feasibility.

04

Emphasize bedroom count

Additional bedrooms lift price and open the property to larger groups and families.

05

Reassess instant booking

Instant booking carries a small negative coefficient. Test whether disabling it supports a higher price point.

06

Fold in the cleaning fee

Guests appear to favor all-inclusive pricing. Test integrating the cleaning fee into the headline price.

07

Segment the market

Shared rooms for budget travelers; private rooms and entire homes for guests who will pay for privacy.

08

Model selected

Polynomial regression with interaction terms: test R^2 0.524, RMSE 0.495 on log price — best of three candidates, no overfitting.

Business problem and solution approach

Problem

- Hosts price by intuition. The objective is a model that predicts listing price from property attributes, so pricing can be set and defended on evidence.
- Target: log price. Evaluation: RMSE, MAE, R^2 , adjusted R^2 , MAPE, on a 70/30 train-test split.

Approach

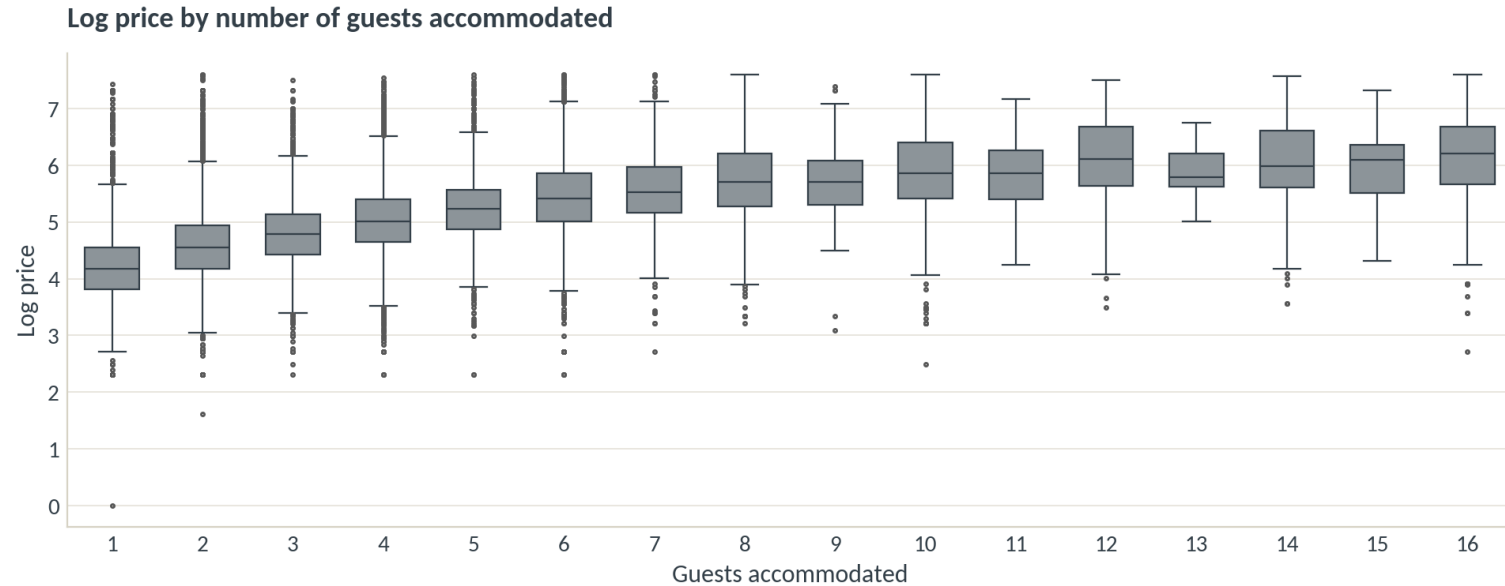
- Data collection and cleaning
- Exploratory analysis — univariate and bivariate
- Preprocessing: duplicates, missing values, outliers, encoding
- Linear regression baseline
- Refinement
 - Polynomial regression (interaction terms only)
 - Forward feature selection
- Evaluation and insight generation

EDA results

What moves price

Log price rises with capacity

Median log price increases steadily with the number of guests accommodated; spread widens at the high end where listings are few.

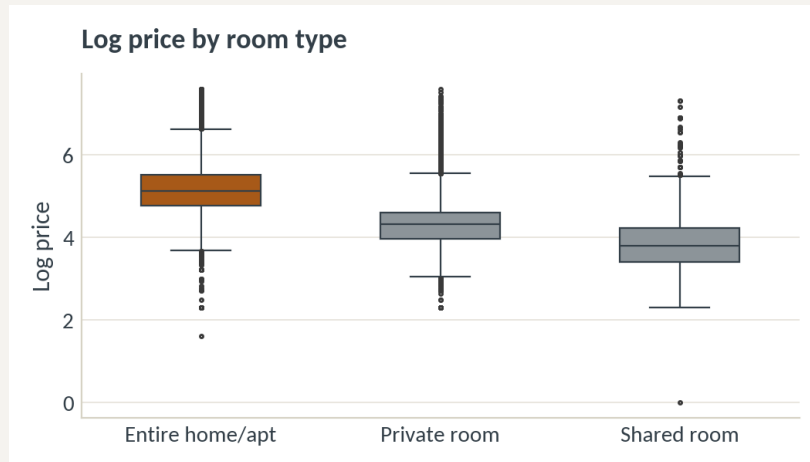


Log price by number of guests accommodated

Room type is the dominant split

Entire homes price highest, shared rooms lowest

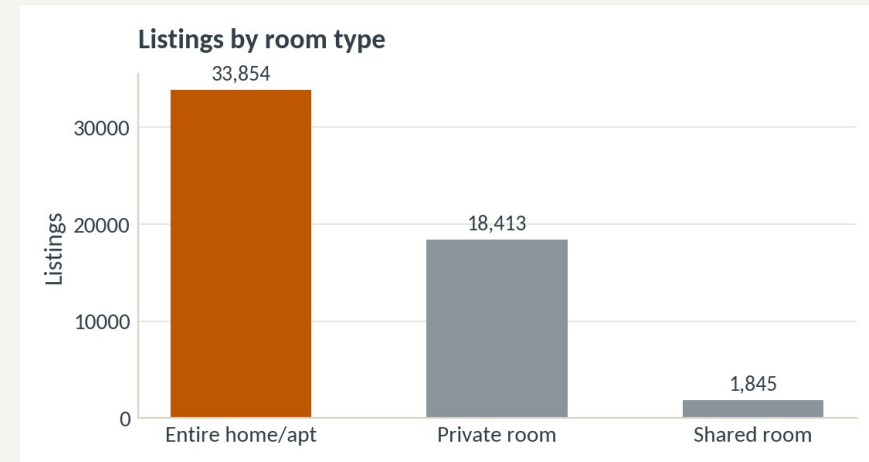
- The three room types barely overlap at the interquartile range.



Log price by room type

Listing mix

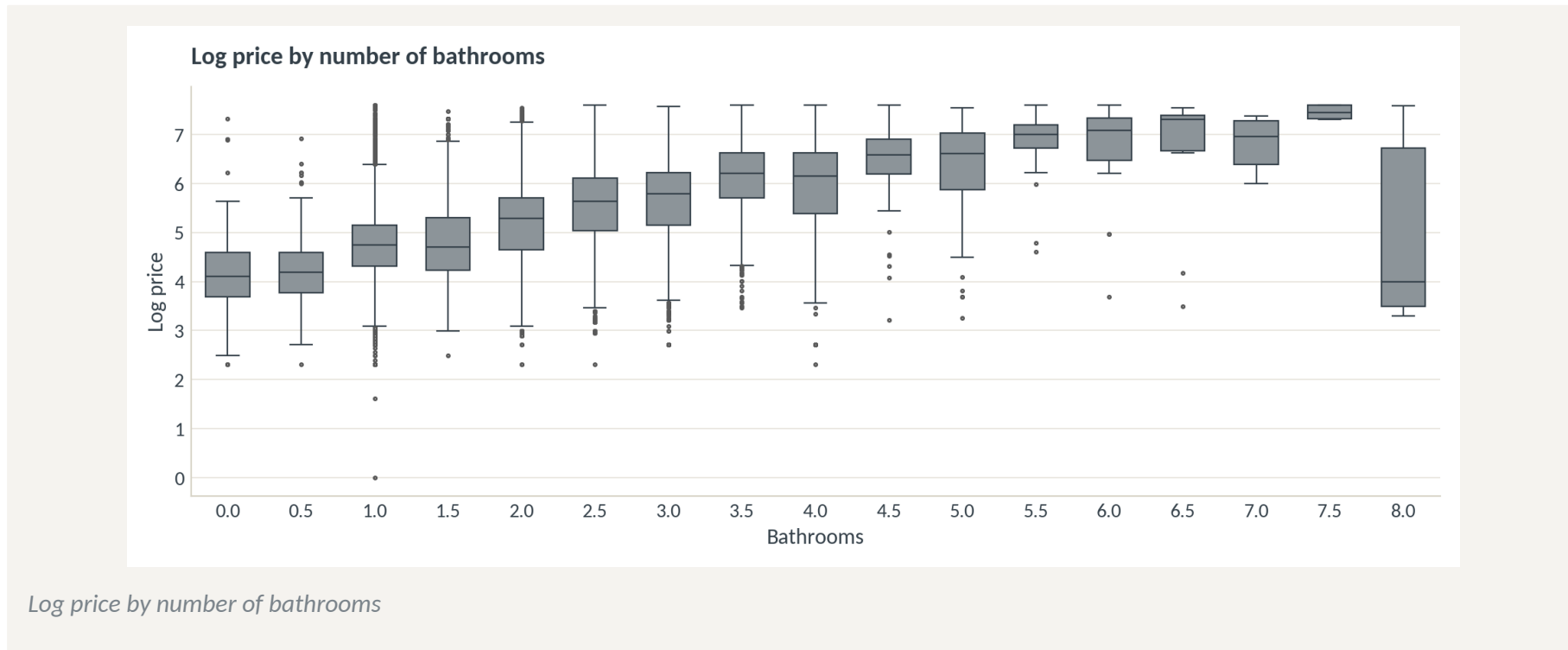
- Entire home/apt 33,854 • Private room 18,413 • Shared room 1,845. Shared rooms are a thin segment.



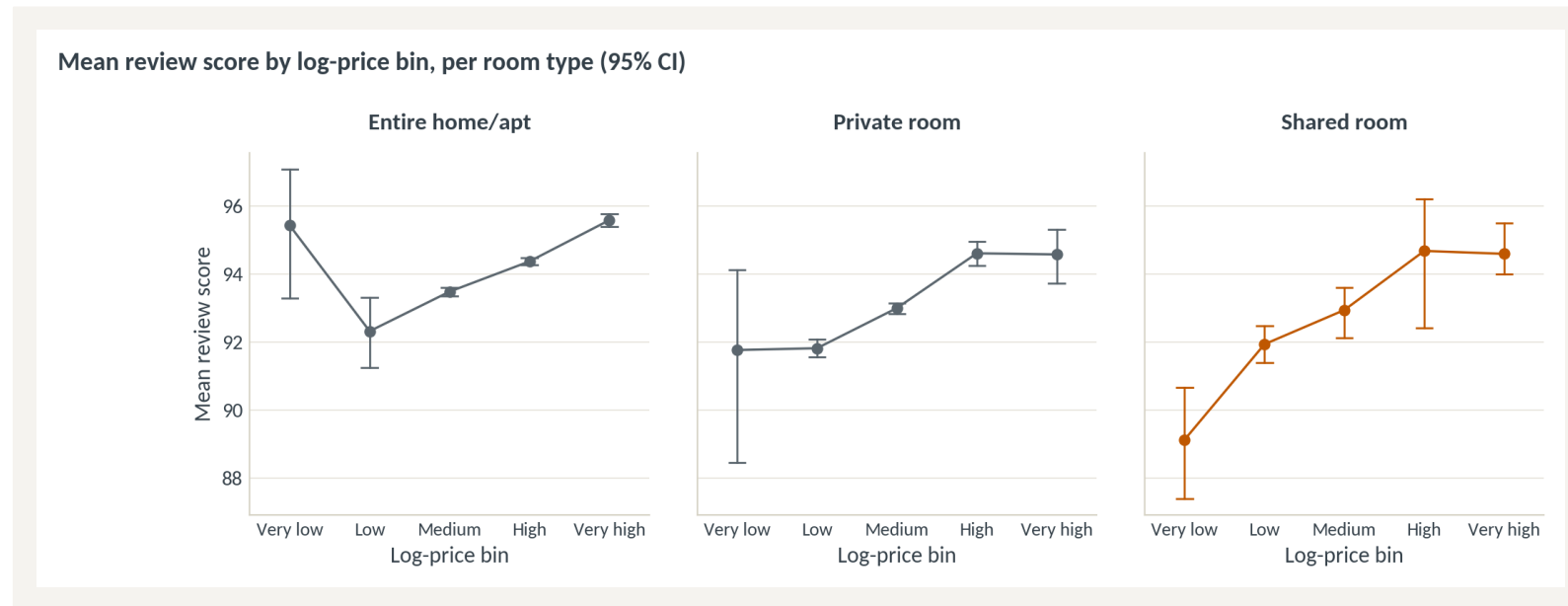
Listings by room type

Bathrooms track price

A general positive relationship between bathroom count and log price, with variance increasing where counts exceed five.



Review scores by price bin and room type



Entire home/apt. Review scores rise with price from very low to very high; variability narrows in the upper bins.

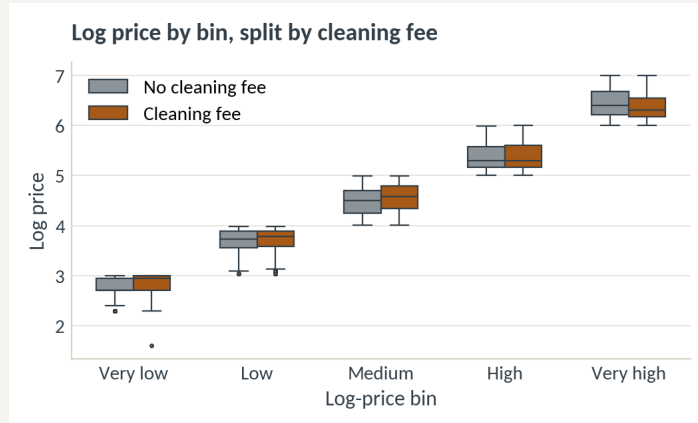
Private room. A weaker upward trend with a dip at medium price; variability roughly constant, wider at the top.

Shared room. The sharpest upward trend, particularly from low to medium; variability is far greater, especially at higher prices.

Cleaning fee and correlation structure

Cleaning fee

- Listings with a cleaning fee price slightly higher in every bin — the fee accompanies, rather than replaces, higher headline price.



Log price by price bin, split by cleaning fee

Numeric correlations

- Accommodates, bedrooms, beds and bathrooms are strongly inter-correlated (0.5–0.7). Log price correlates most with accommodates (0.58) and bedrooms (0.47).



Correlation heatmap, numeric features

Data preprocessing and model preparation

Cleaning

Duplicates. 19,994 dropped from 74,111 rows.

Missing values. Numeric columns imputed with the median grouped by room type; 33 missing prices imputed the same way; 16,722 review scores imputed.

Zero prices. Small constant added to log price for stability.

Outliers. Floored and capped at the whiskers.

Preparation for modeling

- Categorical features encoded; dummy variables for room type and cancellation policy.
- Dependent variable: log price. Dropped log_price_binned from the feature set.
- 70/30 train-test split.

Model performance

Model comparison — train and test

Training set

Test set

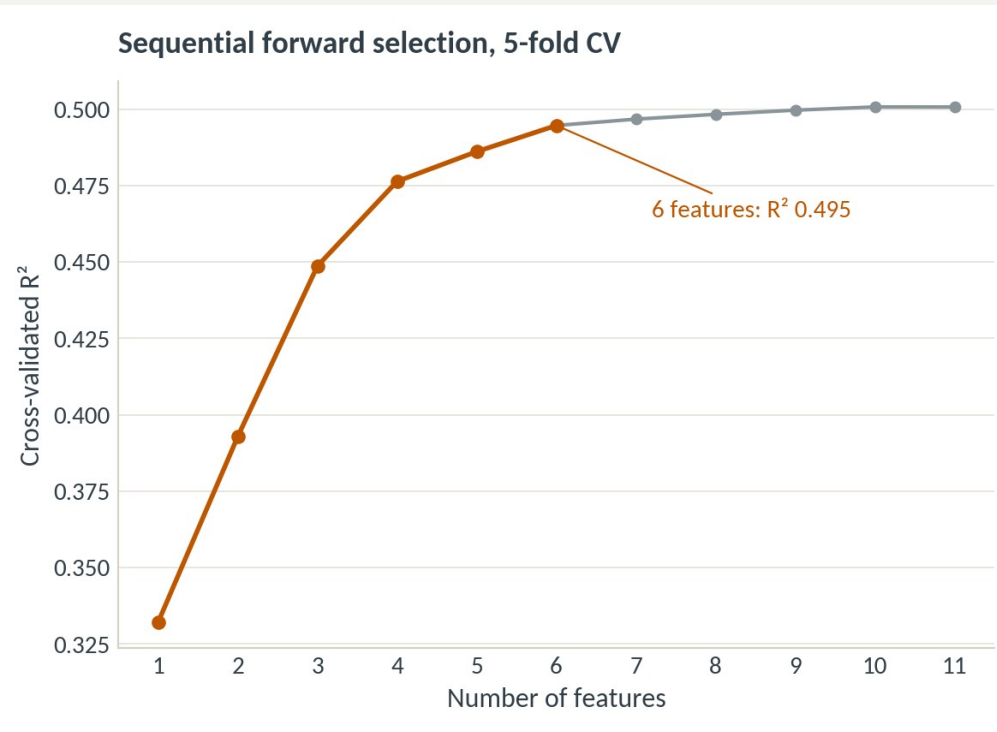
	Linear regression	Polynomial (interaction only)	Linear (SFS features)		Linear regression	Polynomial (interaction only)	Linear (SFS features)
RMSE	0.508	0.498	0.511	RMSE	0.504	0.495	0.507
MAE	0.395	0.387	0.396	MAE	0.392	0.384	0.394
R ²	0.504	0.523	0.499	R ²	0.507	0.524	0.501
Adj. R ²	0.503	0.522	0.499	Adj. R ²	0.506	0.522	0.501
MAPE	8.31	8.13	8.33	MAPE	8.27	8.09	8.31

Polynomial regression (interaction terms only) wins on every metric — lowest RMSE and MAE, highest R² and adjusted R² — and holds that margin from train to test.

The SFS model matches the full linear model to within 0.006 R² using six features instead of eleven — the dropped features carry almost no signal.

Selection. Polynomial regression: marginally better fit, no sign of overfitting (train R² 0.523, test 0.524). It captures interaction effects the linear model cannot — notably room type × bathrooms and bedrooms.

Forward feature selection



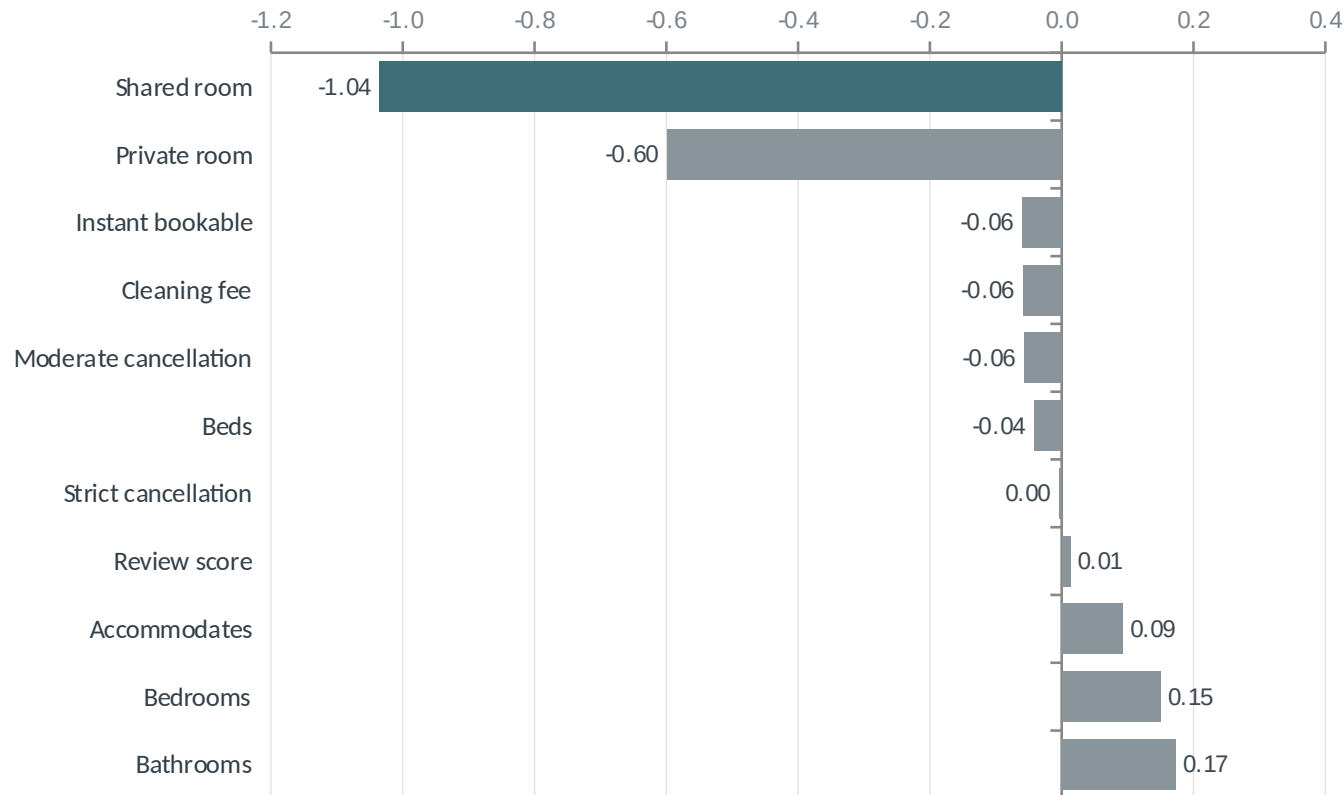
Cross-validated R^2 as features are added

- Sequential forward selection, 5-fold cross-validation, scored on R^2 .
- Performance plateaus after six features; the remaining five add nothing measurable.

Retained: accommodates, bathrooms, review score, bedrooms, room type (both dummies).

Dropped: cleaning fee, instant bookable, beds, cancellation policy. Test R^2 fell from 0.507 to 0.501 — a simpler model at almost no cost.

Feature importance by coefficient (linear model)



Room type dominates. Relative to an entire home, a private room cuts log price by 0.60 and a shared room by 1.04 — the largest effects by an order of magnitude.

Bathrooms, bedrooms, capacity carry the strongest positive coefficients (0.17, 0.15, 0.09).

Cleaning fee and instant booking each shave about 0.06 — small, but both are levers a host controls.

Cancellation policy is close to neutral; strict is indistinguishable from flexible.

Review score has the smallest positive effect: ratings barely move price.

What the interaction terms add

The polynomial model's advantage is that the effect of one feature depends on another. The patterns that matter:

Room type × size. Shared rooms lose further ground with each extra bathroom (−0.54) or bedroom (−0.34); private rooms show the same pattern at half the size. Bedrooms (+0.19) and bathrooms (+0.12) add value for entire homes only.

Cleaning fee. Positive in the polynomial model (+0.21): listings that charge one price higher overall.

Cancellation policy × room type. Moderate policies add a little (+0.13); strict is near-neutral for entire homes but costs shared rooms −0.28.

Instant booking × room type. Positive overall (+0.10), negative for shared rooms (−0.05).

Review score remains the smallest effect (+0.009) in either model.

Appendix

Data background

Data background: 74,111 rows, 11 columns

Dropped 19,994 duplicates. Categorical: room type (3), cancellation policy (3), cleaning fee and instant bookable (boolean).

Variable	Range	Mean / median	Note
log_price	0 – 7.6	4.86 / 4.82	std 0.76; zeros likely special cases or errors
accommodates	1 – 16	3.5 / 3	std 2.34; most listings sleep 2–4
bathrooms	0 – 8	1.30 / 1	zero may denote shared facilities
bedrooms	0 – 10	1.36 / 1	25th and 50th percentiles equal
beds	0 – 18	1.91 / 1	zero could be studio or missing
review_scores_rating	20 – 100	93.2 / 99	75% of listings score 99+; 16,722 imputed



Thank you

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